Bina.az property prices

Decision trees and SVM from scratch

Price prediction and price-tier classification

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MLE-AI-201 / Final project

The dataset and our two tasks

100,775

raw listing rows

64,454

rows after cleaning

TASK A / REGRESSION

Predict the asking price in AZN.
The tree learns log(price).

TASK B / CLASSIFICATION

Premium: price > 210,000 AZN
Standard: price ≤ 210,000 AZN

The tier boundary is the training median. It is not a market-wide luxury label.

Cleaning and input features

Useful inputs

Area, rooms and floors
Location and coordinates
Property type and repair status
Legal and mortgage information

Leakage control

Remove unit_price and total_price.
They reveal the target price.

Exclude names, IDs and free text.

179 model features

Numeric values, missing indicators and one-hot categories

Decision tree: learning useful splits

1. Try a feature and a threshold
2. Choose the largest impurity reduction
3. Repeat until a stopping rule applies

Task	Split criterion	Depth	Min. leaf
Price regression	MSE on log(price)	18	20
Price tier	Gini	10	5

Regression leaves return a mean log price. Class leaves return class counts and probabilities.

Pegasos SVM and the non-linear option

Goal: separate the two price tiers with a wide margin

Loss = λ/2 × ||w||² + mean(max(0, 1 − y × score))

Pegasos update

Learning rate: 1 / (λ × step)
Update weights for margin violations.
Project weights to the allowed size.

Random Fourier Features

Approximate an RBF kernel.
Use 512 cosine features.
Then train the same linear SVM.

Linear: λ = 0.01, 40 epochs RFF: λ = 0.0001, γ = 0.003, 80 epochs

Experiment design and repeatability

Partition	Rows	Purpose
Training	45,117	Fit models and preprocessing
Validation	9,668	Choose settings
Test	9,669	Report final comparisons

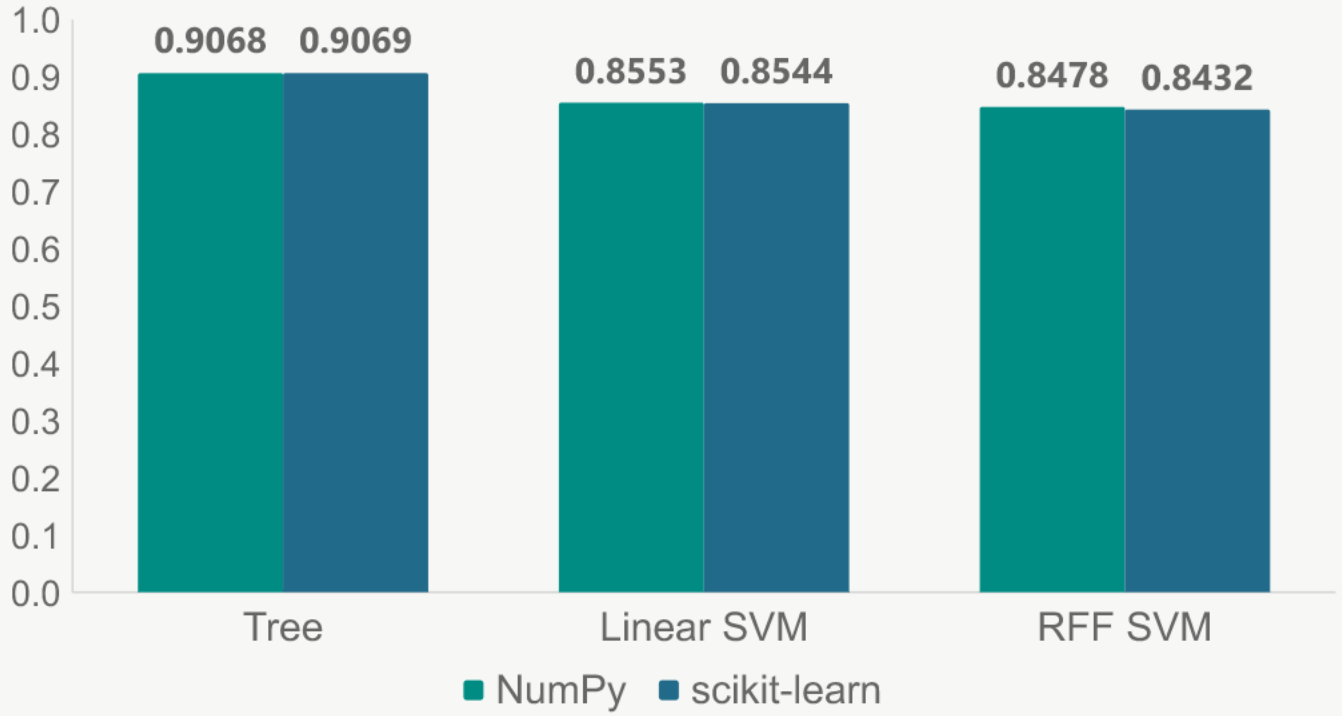
Seed 42

Three-fold CV uses training rows only.
Each fold learns its own preprocessing.

Same data for both implementations

Matched tree settings and SVM inputs.
SVM optimizers differ.

The tree has the strongest classification F1

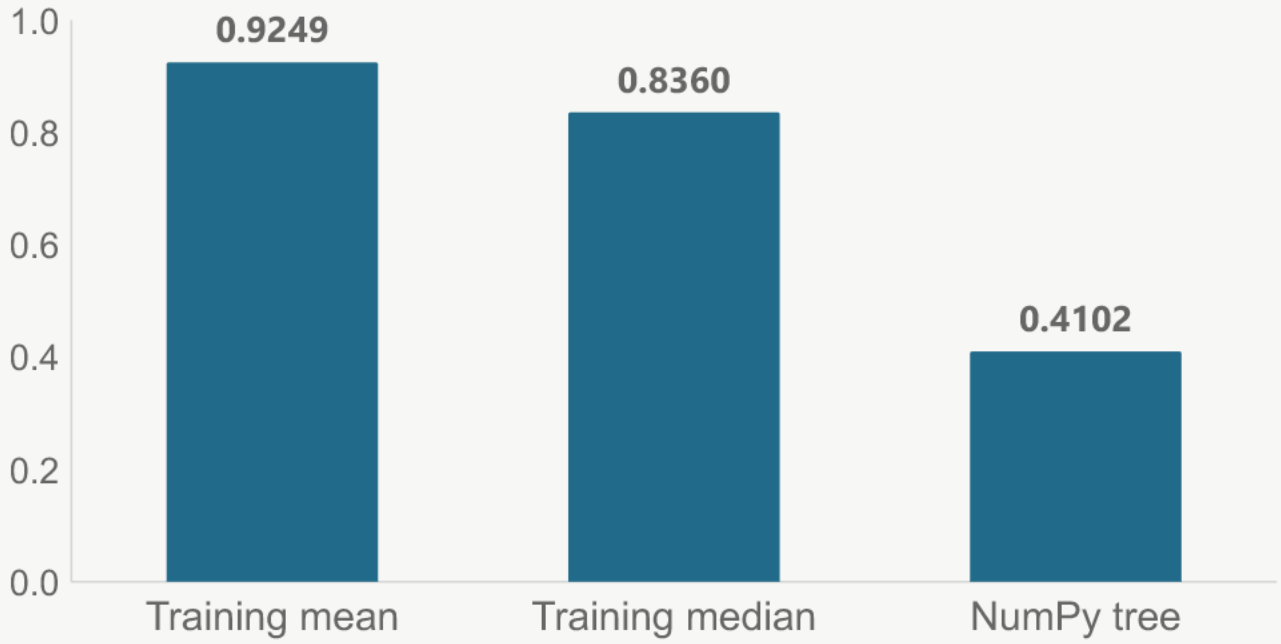


NumPy model	F1	AUC
Tree	0.9068	0.9594
Linear SVM	0.8553	0.9320
RFF SVM	0.8478	0.9275

Test set: 9,669 listings. Higher F1 and ROC-AUC are better.

Regression improves on constant predictions

Test log-RMSE (lower is better)



35.5% lower MAE

Compared with the training-median baseline on the test set.

Test metric	NumPy tree
MAE	185,430 AZN
RMSE	6,184,534 AZN
R^2	0.0105

Validation log-RMSE: 0.3814 for the tree, compared with 0.8035 for the median baseline.

Extreme prices limit the regression result

600,000,000 AZN

A 200 m² apartment in the original dataset

99.97%

of test squared error comes from prices above the training 90th percentile.

What this means

Some source values look suspicious.
We do not know the correct prices.
We keep them in the main results.

Limits: asking prices, possible duplicate properties, and a test set already seen during development.

Team responsibilities and next steps

Yagub Khalilli

Data preparation
Features and leakage checks
End-to-end execution

Present slides 1–3

Ruslan Sadigov

Decision tree
Tree baseline checks
Error and outlier analysis

Present slides 4, 7 and 9

Elmir Esgerov

Pegasos SVM and RFF
Metrics and cross-validation
SVM baseline checks

Present slides 5, 6 and 8

Shared: review each other’s work and explain the full pipeline.

Next: verify unusual listings and evaluate on a new time period.